

Transportation Sustainability and Vehicle Emissions Prediction

A Predictive Analytics Study Using Canadian Fuel Consumption Data

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Abstract

Vehicle carbon dioxide (CO₂) emissions are a major contributor to greenhouse gas emissions and environmental sustainability challenges. Accurate prediction of vehicle emissions can support transportation sustainability analysts, policymakers, and consumers in identifying vehicle characteristics associated with higher emissions and promoting lower-emission transportation choices. This study developed predictive models using the Government of Canada Fuel Consumption Ratings dataset, which contains 10,060 light-duty vehicle records from model years 2015–2024. To reduce target-adjacent proxy leakage, fuel-consumption measures, environmental ratings, and vehicle model were excluded from the final reported models. The final predictor variables included model year, make, vehicle class, engine size, cylinders, transmission, and fuel type.

Two regression models were developed and compared: Multiple Linear Regression as the baseline model and a Tuned Random Forest Regression model as the final machine learning model. Model effectiveness was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R²). Model stability was assessed using five-fold cross-validation, while computational efficiency was evaluated by measuring training and prediction times. The Multiple Linear Regression model achieved a testing R² score of 0.8528, whereas the Tuned Random Forest achieved a testing R² score of 0.9421, demonstrating superior predictive performance. Cross-validation results also showed that the Tuned Random Forest maintained stable performance across different data partitions despite requiring longer training time.

Overall, the Tuned Random Forest was selected as the champion model because it provided the best balance between prediction accuracy, stability, and generalization performance. The findings demonstrate that vehicle specification variables can effectively predict CO₂ emissions without relying on emissions-related proxy variables, providing useful insights for transportation sustainability planning and environmentally informed vehicle comparisons.

1. Introduction

Transportation is one of the largest contributors to greenhouse gas emissions, making the reduction of vehicle carbon dioxide (CO₂) emissions an important environmental objective. Governments, environmental organizations, and the automotive industry continue to promote cleaner transportation technologies and stricter emission regulations to reduce the environmental impact of road transportation. Predictive analytics provides an opportunity to estimate vehicle emissions using historical vehicle data and to identify the vehicle characteristics that contribute most to higher CO₂ emissions. Accurate prediction models can support transportation sustainability analysts in evaluating vehicle emissions, comparing vehicle characteristics, and assisting decision-making related to environmentally responsible transportation planning.

This study develops machine learning models to predict vehicle CO₂ emissions using the Government of Canada Fuel Consumption Ratings dataset containing 10,060 light-duty vehicles from model years 2015–2024. Multiple Linear Regression was selected as the baseline model because of its simplicity and interpretability, while Tuned Random Forest Regression was used to capture more complex relationships among vehicle characteristics. To improve the practical applicability of the models, fuel-consumption measures, environmental ratings, and vehicle model were excluded from the final reported models to reduce the risk of target-adjacent proxy leakage. Model performance was evaluated using prediction accuracy, stability through cross-validation, computational efficiency, and overfitting assessment to identify the most appropriate model for vehicle emissions prediction.

The study addresses the following research questions:

1. Which categories of vehicles sold in Canada between 2015 and 2024 are associated with higher CO₂ emissions, and how can this information support sustainable transportation planning?
2. What vehicle characteristics contribute the most to higher CO₂ emissions in Canadian light-duty vehicles, and how can these insights encourage lower-emission vehicle choices?
3. How can vehicle specifications such as engine size, cylinders, and fuel type help consumers compare more environmentally responsible vehicle options?

2. Literature Review

Vehicle carbon dioxide (CO₂) emissions have become an important research topic because the transportation sector remains one of the largest contributors to greenhouse gas emissions worldwide. Accurate prediction of vehicle emissions can support governments, manufacturers, and environmental organizations in developing strategies to reduce environmental impacts and promote sustainable transportation. Natarajan et al. (2023) reported that vehicle characteristics such as engine size, cylinders, transmission type, and fuel type are strongly associated with CO₂ emissions, making them suitable predictors for emissions modelling. Similarly, Zahid and Jamil (2026) demonstrated that machine learning techniques can effectively estimate vehicle fuel economy and emissions using vehicle specification data, highlighting the growing role of predictive analytics in transportation sustainability.

A common starting point in vehicle emissions prediction is Multiple Linear Regression because of its simplicity, computational efficiency, and interpretability. Regression coefficients provide a clear explanation of how individual vehicle characteristics influence CO₂ emissions, making the model useful as a baseline for comparison. Taşkın et al. (2024) showed that regression-based models provide valuable insights into the relationship between vehicle specifications and emissions but may struggle to represent complex non-linear relationships among variables. Consequently, many vehicle emissions studies use Multiple Linear Regression as an initial benchmark before evaluating more advanced machine learning techniques.

To improve predictive performance, researchers have increasingly adopted ensemble learning methods such as Random Forest Regression. Zahid and Jamil (2026) found that Random Forest was able to capture complex interactions among vehicle characteristics while reducing prediction error compared with traditional regression models. Likewise, Benchmarking Training Emission Prediction Models (2025) reported that ensemble models generally achieved stronger predictive performance because they combined multiple decision trees to model non-linear relationships. In addition to improved prediction accuracy, Random Forest provides feature importance measures that help identify the vehicle characteristics contributing most to emission predictions, making it useful for both prediction and model interpretation.

Although previous studies have demonstrated promising prediction performance, several methodological limitations remain. Many prediction models include fuel-consumption variables or environmental rating variables because they substantially improve prediction accuracy. However, these variables are closely related to vehicle CO₂ emissions and may introduce target-adjacent

proxy leakage, resulting in overly optimistic performance estimates. In addition, most previous studies primarily compare models using prediction accuracy, while comparatively less attention is given to model stability, computational efficiency, overfitting assessment, or the practical implications of the selected features. These differences make it difficult to compare studies directly because datasets, feature selection strategies, and evaluation protocols vary considerably across the literature.

Building on the existing body of knowledge, this study develops predictive models using the Government of Canada Fuel Consumption Ratings dataset containing 10,060 light-duty vehicles from model years 2015–2024. Multiple Linear Regression was selected as the baseline model because it is widely used in previous research and provides an interpretable benchmark for comparison. Random Forest Regression was selected because previous studies have shown that it can effectively model complex, non-linear relationships among vehicle characteristics. To improve the practical applicability of the models, fuel-consumption measures, environmental ratings, and vehicle model were excluded from the final reported models to reduce the risk of target-adjacent proxy leakage. Model performance was evaluated using effectiveness, stability, computational efficiency, and overfitting assessment. The findings of this study are later compared with previous research to determine whether they support existing evidence and to justify the final modelling approach.

3. Dataset and Exploratory Data Analysis

3.1 Dataset Overview

This study uses the Fuel Consumption Ratings dataset published by the Government of Canada (Natural Resources Canada). The dataset contains fuel consumption and carbon dioxide (CO₂) emission information for light-duty vehicles sold in Canada between 2015 and 2024. It includes 10,060 vehicle records and 15 variables consisting of 10 numerical and 5 categorical variables. The target variable selected for this study is CO₂ emissions (g/km), while the remaining variables describe vehicle specifications and characteristics that may influence emissions.

Table 1. Dataset Overview

Dataset Characteristic	Value
Source	Government of Canada (Natural Resources Canada)

Time Period	2015–2024
Total Records	10,060
Total Variables	15
Numerical Variables	10
Categorical Variables	5
Target Variable	CO ₂ Emissions (g/km)

The dataset provides a large and diverse collection of vehicle records, making it suitable for developing and evaluating machine learning regression models. Its combination of numerical and categorical variables supports the comparison of different modelling approaches while providing sufficient information to predict vehicle CO₂ emissions.

3.2 Data Quality Assessment

The dataset was assessed to ensure it was suitable for machine learning analysis. It contained 10,060 records with no duplicate observations, and all variables were stored using appropriate data types. Missing values were present only in the CO₂ Rating and Smog Rating variables, while all other variables were complete.

Table 2. Data Quality Assessment

Quality Check	Observation
Duplicate Records	None
Missing Values	Present only in CO ₂ Rating and Smog Rating
Missing Values in Final Predictors	None
Data Types	Verified
Privacy Issues	None

The selected predictor variables contained no missing values and therefore required no imputation. During feature selection, the CO₂ Rating, Smog Rating, and fuel consumption measures were excluded from the final models because they are closely related to the target variable and could introduce target-adjacent proxy leakage. Overall, the dataset was considered suitable for exploratory data analysis and regression modelling.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to examine the distribution of variables, identify relationships between vehicle characteristics and CO₂ emissions, and detect potential data quality issues before model development. Visualizations including histograms, boxplots, scatter plots, and a correlation heatmap were used to understand the dataset and support feature selection.

Figure 1. Distribution of CO₂ Emissions

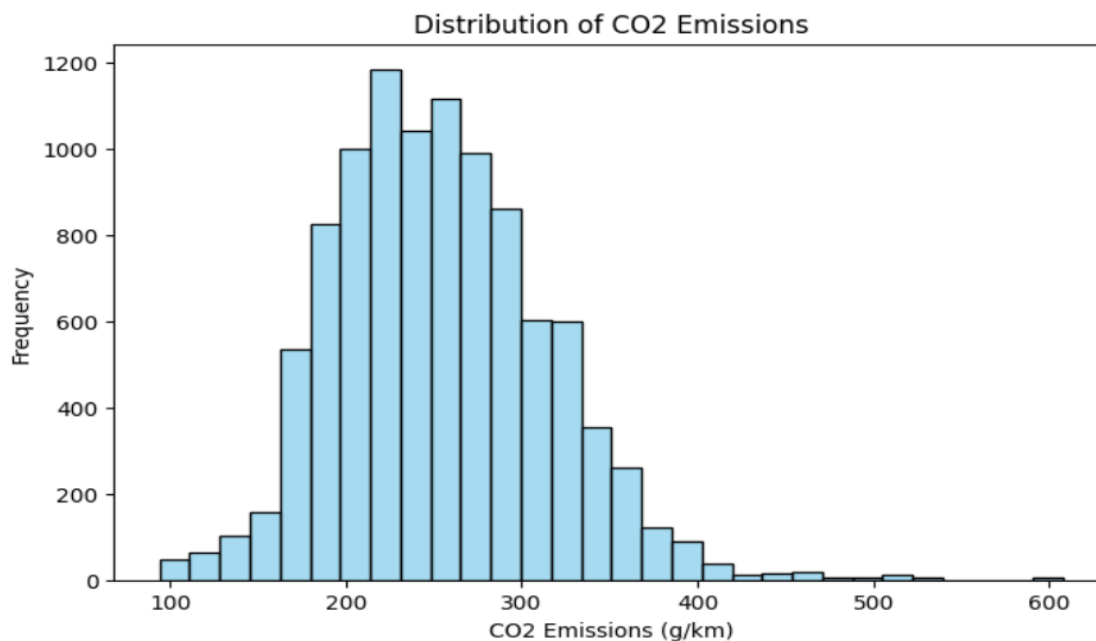
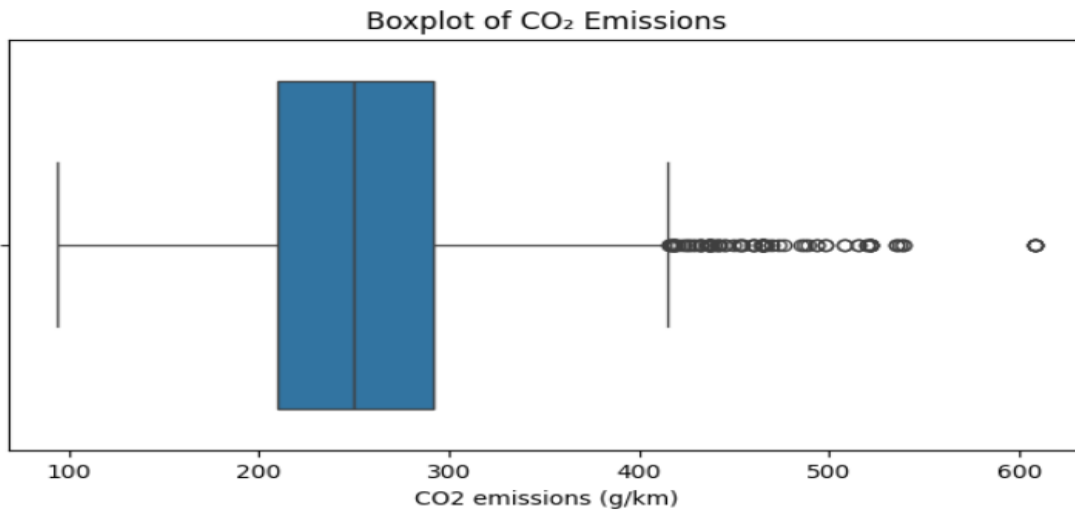


Figure 1 shows that most vehicles produce between 180 and 300 g/km of CO₂ emissions, with fewer vehicles exhibiting very low or very high emission levels. The distribution is moderately right-skewed, indicating the presence of some high-emission vehicles while maintaining sufficient variability for regression modelling.

To further examine the distribution of CO₂ emissions and identify potential extreme observations, a boxplot was generated.

Figure 2. Boxplot of CO₂ Emissions



The boxplot identifies several high-emission outliers. These observations represent vehicles with larger engines and higher fuel consumption rather than data errors, so they were retained to preserve realistic variation within the dataset.

The relationship between engine size and CO₂ emissions was investigated to determine whether larger engines are associated with increased emissions.

Figure 3. Relationship Between Engine Size and CO₂ Emissions

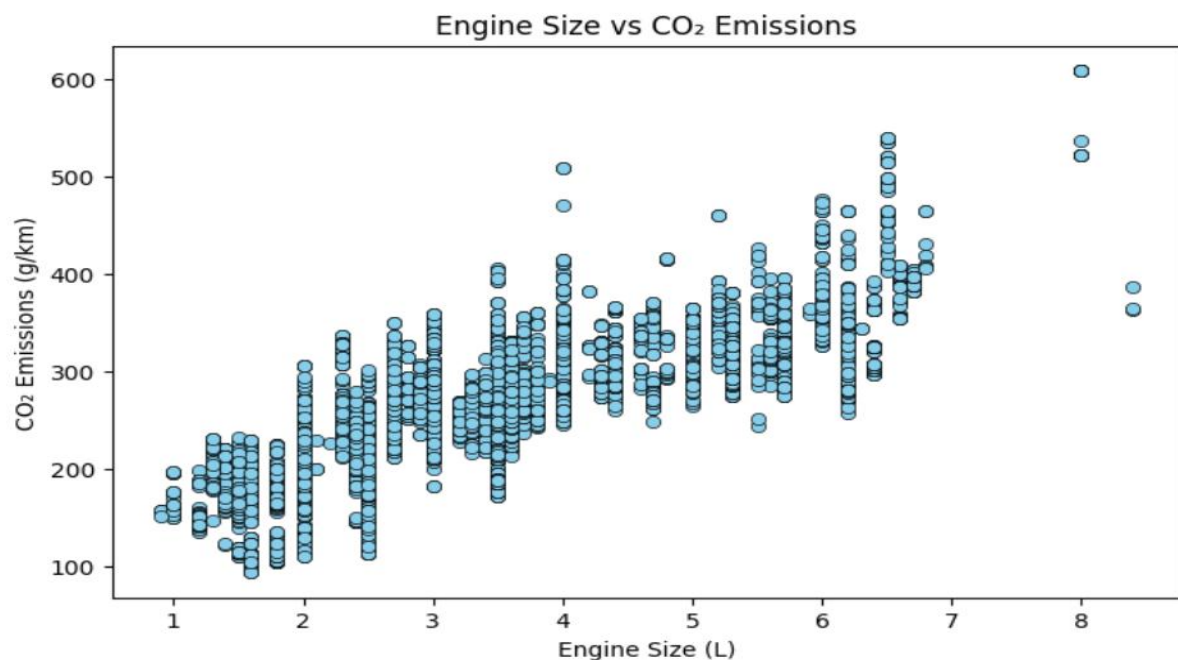
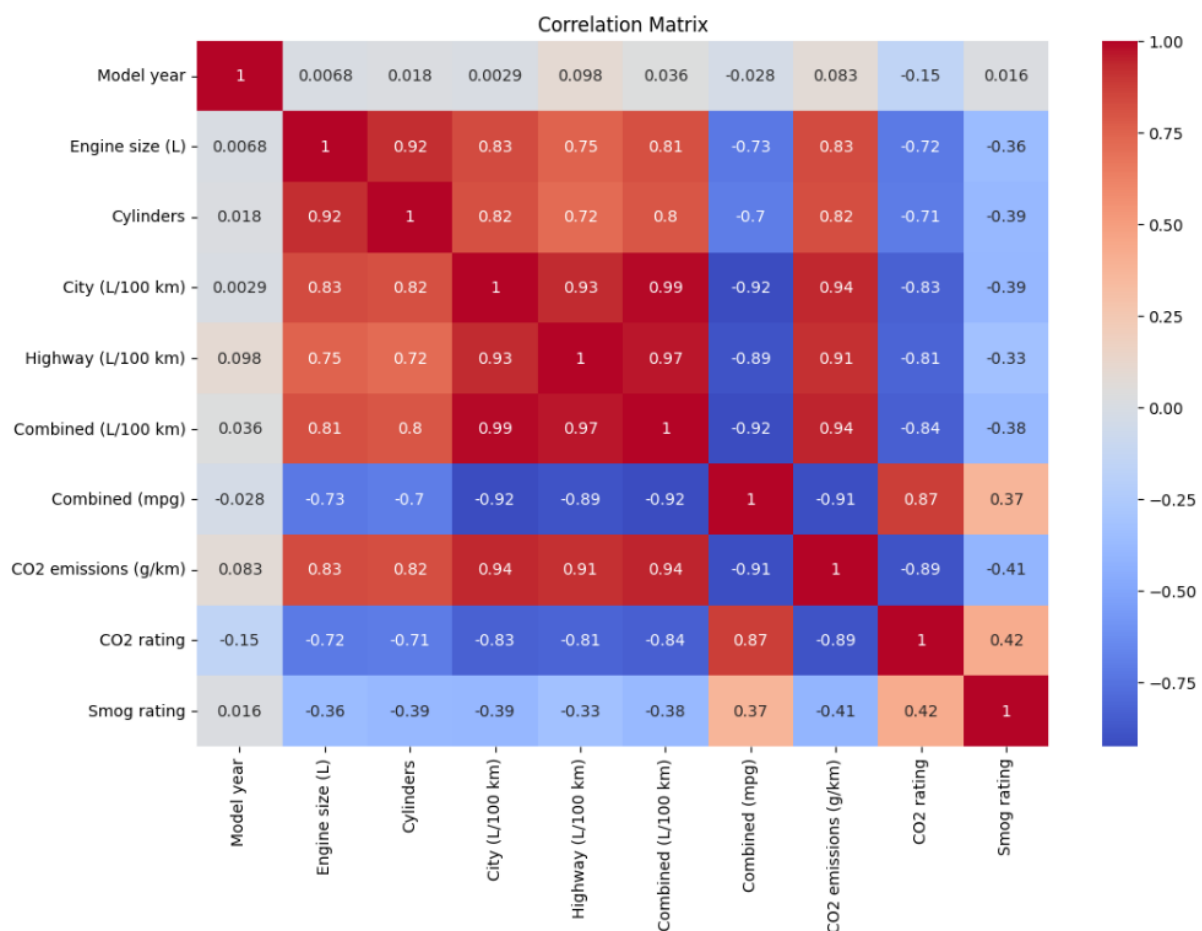


Figure 3 demonstrates a positive relationship between engine size and CO₂ emissions. Vehicles with larger engines generally produce higher emissions, although some variation exists due to other vehicle characteristics. This relationship supported the inclusion of engine size as an important predictor.

A correlation heatmap was created to evaluate the strength of linear relationships among the numerical variables and identify variables that may introduce target leakage.

Figure 4. Correlation Heatmap



The correlation heatmap indicates strong positive relationships between CO₂ emissions, engine size, and cylinders. Fuel-consumption variables also show very high correlations with the target variable; therefore, they were excluded from the final models to reduce target-adjacent proxy leakage and improve model validity.

3.4 Summary of Exploratory Data Analysis

The exploratory analysis confirmed that the dataset was suitable for regression modelling. Engine size and cylinders demonstrated strong relationships with CO₂ emissions, while the visualizations identified no data quality issues requiring the removal of observations. The findings also supported excluding fuel-consumption variables from the final models because of their close relationship with the target variable.

4. Methodology

4.1 Overview

This chapter describes the methodology used to develop and evaluate the machine learning models for predicting vehicle CO₂ emissions. The workflow included feature selection, data preprocessing, model development, and model evaluation. All preprocessing and modelling steps were performed using Python in Google Colab with libraries including pandas, NumPy, scikit-learn, matplotlib, and seaborn. To ensure reproducibility and avoid data leakage, the training and testing datasets were separated before any preprocessing operations that required learning from the data.

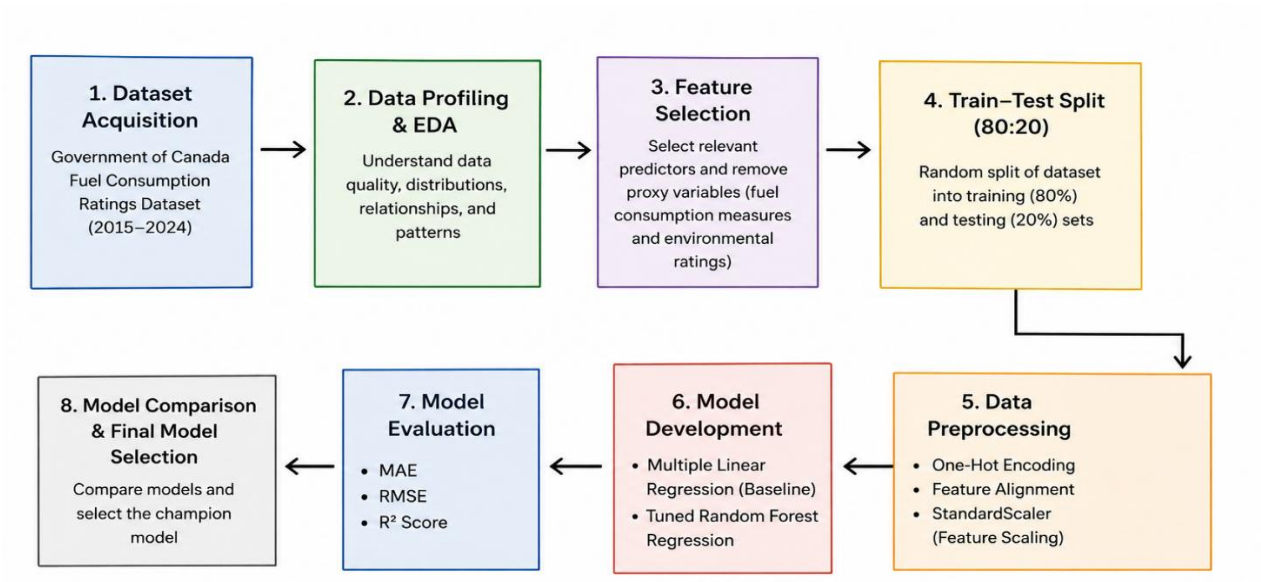
4.2 Project Methodology

The project followed a structured workflow beginning with data acquisition from the Government of Canada Fuel Consumption Ratings dataset. The data were first examined through data profiling and exploratory data analysis to understand their quality, structure, and relationships among variables. Based on the findings, appropriate predictor variables were selected and unnecessary variables were removed.

The cleaned dataset was divided into training and testing sets before preprocessing. Categorical variables were encoded using One-Hot Encoding, while numerical variables were standardized using StandardScaler. Two regression models, Multiple Linear Regression and Random Forest Regression, were then developed and evaluated using the testing dataset. Finally, the models were compared using multiple evaluation metrics to identify the most suitable approach for predicting vehicle CO₂ emissions.

Figure 5 summarises the complete workflow followed throughout this study, from data acquisition to final model selection.

Figure 5. Overall Project Workflow



4.3 Feature Selection

Feature selection was performed to retain variables that contribute meaningful predictive information while reducing the risk of target-adjacent proxy leakage. Based on the exploratory data analysis and project objectives, the final predictor variables included Model Year, Make, Vehicle Class, Engine Size (L), Cylinders, Transmission, and Fuel Type. These variables represent vehicle characteristics that are available before measuring emissions and are therefore appropriate predictors for estimating CO₂ emissions.

Table 3. Final Feature Selection

Variable Type	Variables
Predictor Variables	Model Year, Make, Vehicle Class, Engine Size, Cylinders, Transmission, Fuel Type
Target Variable	CO ₂ Emissions (g/km)

Variables directly related to the target variable, including Fuel Consumption (City, Highway, Combined, and Combined MPG), CO₂ Rating, and Smog Rating, were excluded from the final

models because they are derived from or closely associated with CO₂ emissions. Removing these variables reduces circular prediction and improves the validity of model evaluation.

4.4 Data Preprocessing and Model Development

The dataset was divided into 80% training data and 20% testing data before applying any fitted preprocessing steps. This order was used to prevent information from the testing data from influencing model development. The selected numerical variables, including Model Year, Engine Size, and Cylinders, were standardised using StandardScaler. Categorical variables, including Make, Vehicle Class, Transmission, and Fuel Type, were converted into numerical form using one-hot encoding. The training and testing datasets were then aligned to ensure that both contained the same feature columns.

Two regression models were developed using the processed data. Multiple Linear Regression was used as the baseline model because it provides a simple and interpretable method for estimating the relationship between vehicle characteristics and CO₂ emissions. Random Forest Regression was selected as the comparison model because it can capture non-linear relationships and interactions among predictors. Both models were trained on the training dataset and evaluated using the unseen testing dataset.

4.5 Model Development

The developed models were evaluated using the testing dataset to compare their prediction performance. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) were used as the primary evaluation metrics because they provide complementary information about prediction accuracy.

To further assess model reliability, 5-fold cross-validation was performed on both models to examine the consistency of their performance across different subsets of the training data. Training time and prediction time were also recorded to compare the computational efficiency of each model. The combined results from these evaluations were used to identify the model that achieved the best balance between prediction accuracy, stability, and efficiency.

4.6 Hyperparameter Tuning

Random Forest Regression hyperparameters were optimised using GridSearchCV with five-fold cross-validation to improve model performance and reduce overfitting. The search evaluated

multiple combinations of the number of trees (`n_estimators`), maximum tree depth (`max_depth`), minimum samples required to split a node (`min_samples_split`), and minimum samples required at a leaf node (`min_samples_leaf`). The parameter combination with the lowest cross-validation Root Mean Squared Error (RMSE) was selected for the final Random Forest model.

The hyperparameter tuning process identified the combination of parameters that produced the lowest cross-validation RMSE. The resulting GridSearchCV output is presented in Figure 6.

Figure 6. Random Forest Hyperparameter Tuning Results

```
*** Fitting 5 folds for each of 16 candidates, totalling 80 fits
Best Parameters:
{'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators': 200}

Best Cross-Validation RMSE:
15.55070040499395
```

The selected configuration balances model complexity and predictive performance. Using 200 decision trees improves prediction stability by averaging the results of multiple trees. Limiting the maximum tree depth to 20 allows the model to capture complex relationships while reducing the likelihood of overfitting. Setting `min_samples_split = 5` prevents unnecessary splits based on very small sample sizes, whereas `min_samples_leaf = 1` allows terminal nodes to retain detailed information when supported by the data. The tuned Random Forest model was subsequently used for final model evaluation and comparison with the baseline Multiple Linear Regression model.

4.7 Model Evaluation Metrics

The predictive performance of both regression models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MAE measures the average prediction error, RMSE gives greater weight to larger prediction errors, and R^2 indicates the proportion of variation in CO₂ emissions explained by the model

Table 4. Regression Evaluation Metrics

Evaluation Method	Purpose
Mean Absolute Error (MAE)	Measures average prediction error
Root Mean Squared Error (RMSE)	Measures prediction error while giving greater weight to larger errors
R ² Score	Measures the proportion of variance explained by the model
Five-Fold Cross-Validation	Evaluates model stability and generalization
Training Time	Measures computational cost during model training
Prediction Time	Measures computational efficiency during prediction
Feature Importance	Identifies the most influential predictor variables in the Random Forest model

These metrics provide complementary measures of model accuracy and were used to compare the predictive performance of Multiple Linear Regression and Random Forest Regression

5. Model Results and Evaluation

This section evaluates the performance of the developed models for predicting vehicle CO₂ emissions. The results obtained from Multiple Linear Regression and Random Forest Regression are evaluated using regression performance metrics, graphical analysis, and model validation techniques. The findings from both models are then compared to determine the most suitable approach for predicting vehicle CO₂ emissions and to address the research objectives of this study.

5.1 Multiple Linear Regression Results

Multiple Linear Regression was developed as the baseline model to predict vehicle CO₂ emissions using the selected vehicle specification variables. After training the model with the preprocessed training dataset, predictions were generated for the testing dataset and evaluated using regression performance metrics. The baseline model provides a reference for assessing the performance improvements achieved by the Random Forest Regression model.

5.1.1 Model Performance

The performance metrics obtained from the Multiple Linear Regression model are presented in Figure 7.

Figure 7. Multiple Linear Regression Evaluation Metrics

MAE: 17.091308793478277
RMSE: 22.888462574285928
R2 Score: 0.8528172086252866

Figure 7 presents the evaluation metrics obtained for the Multiple Linear Regression model. The model achieved a Mean Absolute Error (MAE) of 17.09 g/km, indicating that the predicted CO₂ emission values differed from the observed values by approximately 17 grams per kilometre on average. The Root Mean Squared Error (RMSE) of 22.89 g/km shows that larger prediction errors had a greater influence on the overall model error. The model also achieved an R² score of 0.8528, indicating that approximately 85% of the variation in vehicle CO₂ emissions was explained by the selected predictor variables.

Overall, the results demonstrate that Multiple Linear Regression provides a strong baseline for predicting vehicle CO₂ emissions. However, the remaining unexplained variation suggests that the relationships between vehicle characteristics and emissions are not entirely linear and may be better captured by more flexible machine learning algorithms.

5.1.2 Model Diagnostics

Model diagnostics were performed to examine how closely the predicted CO₂ emission values matched the actual values in the testing dataset. A predicted-versus-actual plot was used to assess the overall agreement between observed and predicted emissions and to identify whether prediction errors increased at higher emission levels.

Figure 8. Predicted versus Actual CO₂ Emissions

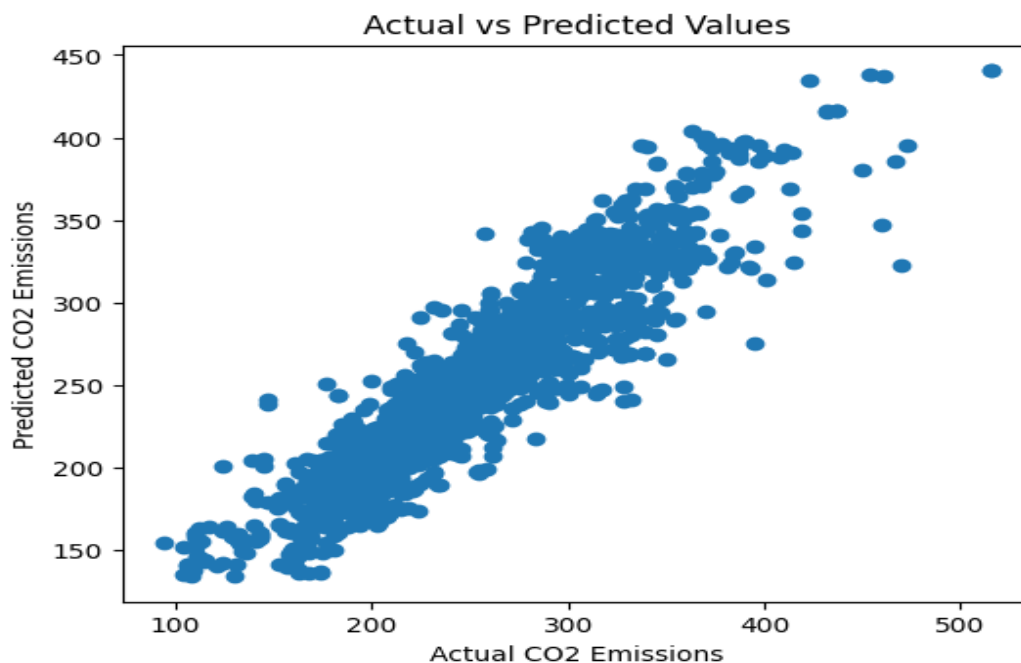


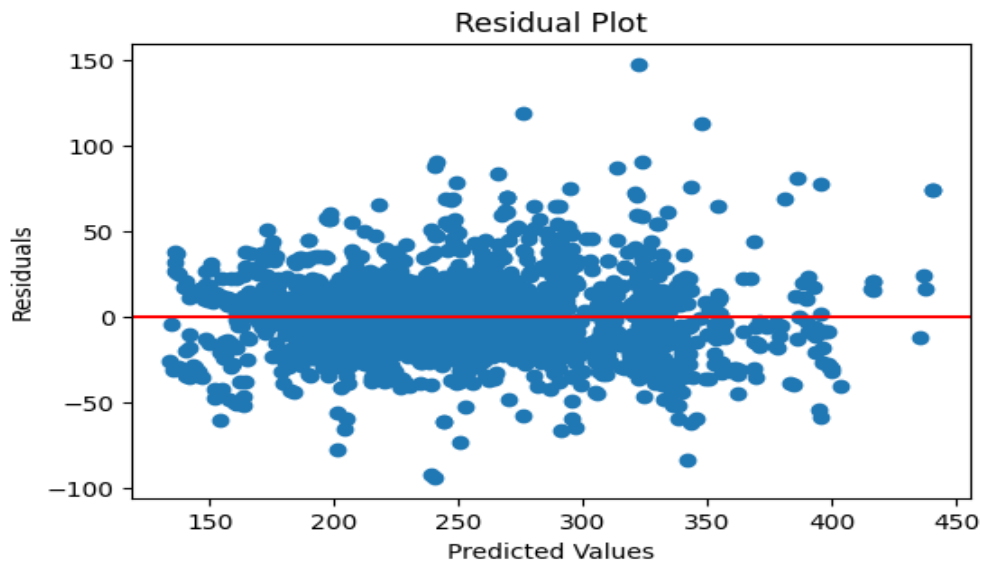
Figure 8 compares the predicted CO₂ emission values with the corresponding observed values in the testing dataset. Most observations are concentrated near the diagonal reference line, indicating that the model successfully captures the overall relationship between vehicle characteristics and CO₂ emissions.

Although the majority of predictions closely match the actual values, a wider spread of observations is visible for vehicles with higher CO₂ emissions. This suggests that the model produces larger prediction errors for vehicles with extreme emission values. Nevertheless, the overall distribution demonstrates that the model provides reliable predictions across a broad range of vehicle types.

5.1.3 Residual Analysis

The residual plot shows that prediction errors are generally distributed around zero without a clear systematic pattern. This indicates that the model captures the overall relationship between the predictor variables and the target variable reasonably well, although larger residuals are visible for some high-emission vehicles. Overall, the residual distribution suggests that the model does not exhibit substantial systematic bias and provides reasonably consistent predictions across most observations.

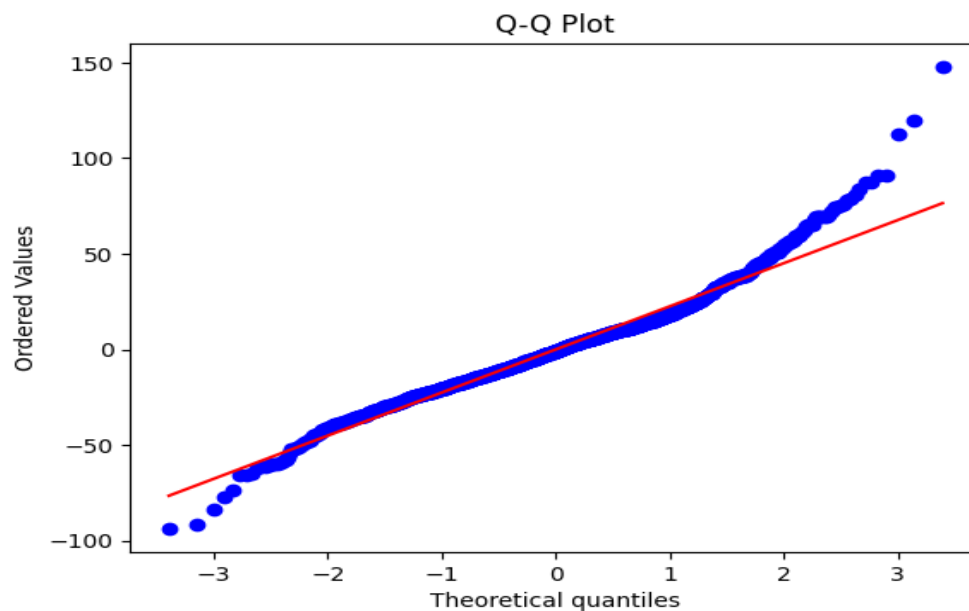
Figure 9. Residual Plot for Multiple Linear Regression



5.1.4 Assumption Assessment

To assess the normality assumption of Multiple Linear Regression, a Q-Q plot of the residuals was examined. This plot compares the distribution of the model residuals with the theoretical normal distribution and helps determine whether the residuals follow an approximately normal pattern.

Figure 10. Q-Q Plot of Residual



The regression assumptions were assessed using residual analysis and multicollinearity diagnostics. The residual distribution is centred near zero, while the VIF values indicate that multicollinearity among the predictor variables remains within acceptable limits. These findings support the suitability of the Multiple Linear Regression model for this analysis.

5.2 Random Forest Regression Results

The Random Forest Regression model was evaluated using the same testing dataset to determine whether a non-linear machine learning approach could improve prediction accuracy compared with the baseline model.

5.2.1 Model Performance

The Random Forest model achieved lower prediction errors and a higher R^2 score than the Multiple Linear Regression model, demonstrating superior predictive performance. These results indicate that the ensemble learning approach is more effective at modelling the complex relationships within the dataset.

Figure 11. Random Forest Evaluation Metrics

MAE: 9.627359956017445
RMSE: 14.507343281483367
R2 Score: 0.9408711317188369

Figure 11 presents the performance metrics obtained for the Random Forest Regression model. Compared with the Multiple Linear Regression model, the Random Forest achieved lower prediction errors and a higher R^2 score, demonstrating improved predictive performance. These results indicate that the ensemble model more effectively captured the complex relationships between vehicle characteristics and CO₂ emissions.

The improved performance is consistent with the ability of Random Forest Regression to model non-linear relationships and interactions among predictor variables. Consequently, the Random Forest model produced more accurate predictions across the testing dataset.

5.2.2 Model Generalization

Model generalization was assessed by comparing the Random Forest model's performance on the training and testing datasets. Similar performance across both datasets indicates that the model generalizes well to unseen data and is not substantially overfitted.

Figure 12. Training and Testing Performance

```
Tuned Random Forest Results

Training Performance
MAE: 7.713786296928301
RMSE: 11.578878119538556
R2 Score: 0.9635704353285957

Testing Performance
MAE: 9.776943669468837
RMSE: 14.355533850376036
R2 Score: 0.942102143373654
```

Figure 12 shows that the Tuned Random Forest achieved a training R^2 of 0.9636 and a testing R^2 of 0.9421. The relatively small difference of approximately 0.0215 indicates that the model performed similarly on training and unseen testing data. The testing MAE and RMSE were moderately higher than the training errors, which is expected, but the differences were not large enough to indicate substantial overfitting. Therefore, the model demonstrates good generalisation performance.

5.3 Model Comparison

Table 5. Performance Comparison of Developed Models

Evaluation Metric	Multiple Linear Regression	Tuned Random Forest	Better Model
Mean Absolute Error (MAE) (g/km)	17.09	9.78	Tuned Random Forest

Root Mean Squared Error (RMSE) (g/km)	22.89	14.36	Tuned Random Forest
Coefficient of Determination (R^2)	0.8528	0.9421	Tuned Random Forest

Table 5 compares the predictive performance of the Multiple Linear Regression and Tuned Random Forest models using the three-evaluation metrics. Both models produced reliable predictions; however, the Tuned Random Forest consistently outperformed the Multiple Linear Regression model across all evaluation measures.

The Multiple Linear Regression model achieved an MAE of 17.09 g/km, an RMSE of 22.89 g/km, and an R^2 score of 0.8528. In comparison, the Tuned Random Forest reduced the MAE to 9.78 g/km and the RMSE to 14.36 g/km, while increasing the R^2 score to 0.9421. These improvements indicate that the Random Forest model generated more accurate predictions and explained a greater proportion of the variation in vehicle CO₂ emissions.

The improved performance can be attributed to the ability of Random Forest Regression to capture non-linear relationships and interactions among vehicle characteristics that are difficult to represent using a linear regression model. Based on these results, the Tuned Random Forest was selected as the final prediction model for this study because it achieved the highest prediction accuracy while maintaining good generalization performance.

6 Multi-Angle Model Evaluation

6.1 Prediction Effectiveness

Prediction effectiveness refers to how accurately each model estimated vehicle CO₂ emissions using the testing dataset. As demonstrated in Chapter 5, both the Multiple Linear Regression and Tuned Random Forest models produced reliable predictions. However, the Tuned Random Forest consistently achieved lower prediction errors and a higher coefficient of determination than the baseline model.

Model effectiveness refers to the ability of each model to accurately predict CO₂ emissions on unseen data. This comparison is based on MAE, RMSE, and R^2 .

Table 6. Prediction Effectiveness

Evaluation Metric	Multiple Linear Regression	Tuned Random Forest	Better Model
MAE (g/km)	17.09	9.78	Tuned Random Forest
RMSE (g/km)	22.89	14.36	Tuned Random Forest
R ² Score	0.8528	0.9421	Tuned Random Forest
Generalization	Good	Very Good	Tuned Random Forest
Overall Performance	Good	Excellent	Tuned Random Forest

The Multiple Linear Regression model achieved an MAE of 17.09 g/km, an RMSE of 22.89 g/km, and an R² score of 0.8528. In comparison, the Tuned Random Forest reduced the MAE to 9.78 g/km and the RMSE to 14.36 g/km, while increasing the R² score to 0.9421. These improvements indicate that the Random Forest model generated predictions that were consistently closer to the observed CO₂ emission values.

The superior performance of the Tuned Random Forest demonstrates its ability to capture complex relationships among vehicle characteristics that cannot be fully represented using a linear model. Consequently, the Random Forest model provided more accurate predictions across the full range of vehicles included in the dataset.

6.2 Model Stability and Efficiency

Model stability was evaluated using five-fold cross-validation to determine how consistently each model performed across different subsets of the training data. Table 7 presents the mean cross-validation Root Mean Squared Error (RMSE) and the corresponding standard deviation for both models. The Multiple Linear Regression model achieved a mean cross-validation RMSE of 22.77 g/km with a standard deviation of 0.39, whereas the Tuned Random Forest achieved a lower mean cross-validation RMSE of 15.55 g/km with a standard deviation of 0.68. The relatively small standard deviations indicate that both models produced consistent performance across the five

folds, demonstrating good generalization. However, the Tuned Random Forest consistently achieved lower prediction errors than the Multiple Linear Regression model, indicating superior predictive performance while maintaining reliable performance across different data partitions

Table 7. Five-Fold Cross-Validation Results

Model	Mean CV RMSE (g/km)	Standard Deviation
Multiple Linear Regression	22.77	0.39
Tuned Random Forest	15.55	0.68

Figure 13 compares the computational efficiency of the two models. Multiple Linear Regression required only **0.0513 seconds** for training and **0.0037 seconds** for prediction, making it computationally inexpensive. In contrast, the Tuned Random Forest required **7.4933 seconds** for training and **0.1035 seconds** for prediction because it constructs and evaluates multiple decision trees. Although the Tuned Random Forest required more computational time, the additional cost was acceptable given its substantially better predictive performance, including a higher R^2 score and lower MAE and RMSE. Therefore, the improvement in prediction accuracy justified the increased computational complexity for this vehicle emissions prediction application.

Figure 13. Efficiency Comparison of the Two Models

	Model	Training Time (seconds)	Prediction Time (seconds)
0	Multiple Linear Regression	0.0513	0.0037
1	Tuned Random Forest	7.4933	0.1035

6.3 Champion Model Selection

Based on the combined evidence from prediction effectiveness, model stability, and computational efficiency, the Tuned Random Forest was selected as the champion model. It achieved a lower MAE of 9.78 g/km and RMSE of 14.36 g/km and a higher R^2 of 0.9421 than Multiple Linear Regression. It also maintained similar training and testing performance and stable results across five-fold cross-validation. Although it required more training and prediction time, the additional computational cost remained acceptable for this offline vehicle-emissions analysis.

Although Random Forest is more complex than Multiple Linear Regression, it provides valuable insights through feature importance analysis. The results showed that Engine Size and Cylinders were the most influential predictors of CO₂ emissions, followed by Vehicle Class, Fuel Type, Make, and Model Year. These findings are consistent with the exploratory data analysis and support the interpretation that larger engines and vehicles with more cylinders generally produce higher emissions. The identified relationships can assist consumers in comparing vehicle options, support sustainability analysts in identifying high-emission vehicle categories and provide policymakers with evidence for developing transportation and environmental strategies. Therefore, the Random Forest model offers both strong predictive performance and practical value for supporting emission-related decision-making.

7. Research Question Findings and Stakeholder Impact

7.1 Overview

This chapter discusses how the findings obtained from the machine learning models addressed the research questions established at the beginning of this study. The results are interpreted in relation to vehicle CO₂ emissions, key influencing vehicle characteristics, and the practical applications of the developed prediction model. The chapter also highlights the potential benefits of the proposed approach for various stakeholders involved in transportation sustainability and environmental decision-making.

7.2 Research Question 1

Which categories of vehicles sold in Canada between 2015 and 2024 are associated with higher CO₂ emissions, and how can this information support sustainable transportation planning?

The analysis indicates that larger vehicle categories generally produce higher CO₂ emissions than smaller vehicle classes. Passenger vans, standard SUVs, standard pickup trucks, and two-seater performance vehicles exhibited the highest emission levels, whereas compact cars, small station wagons, and special purpose vehicles generally produced lower emissions. This suggests that vehicle class is an important factor influencing emissions and should be considered when evaluating transportation sustainability.

Figure 14. CO₂ emissions vs vehicle classes

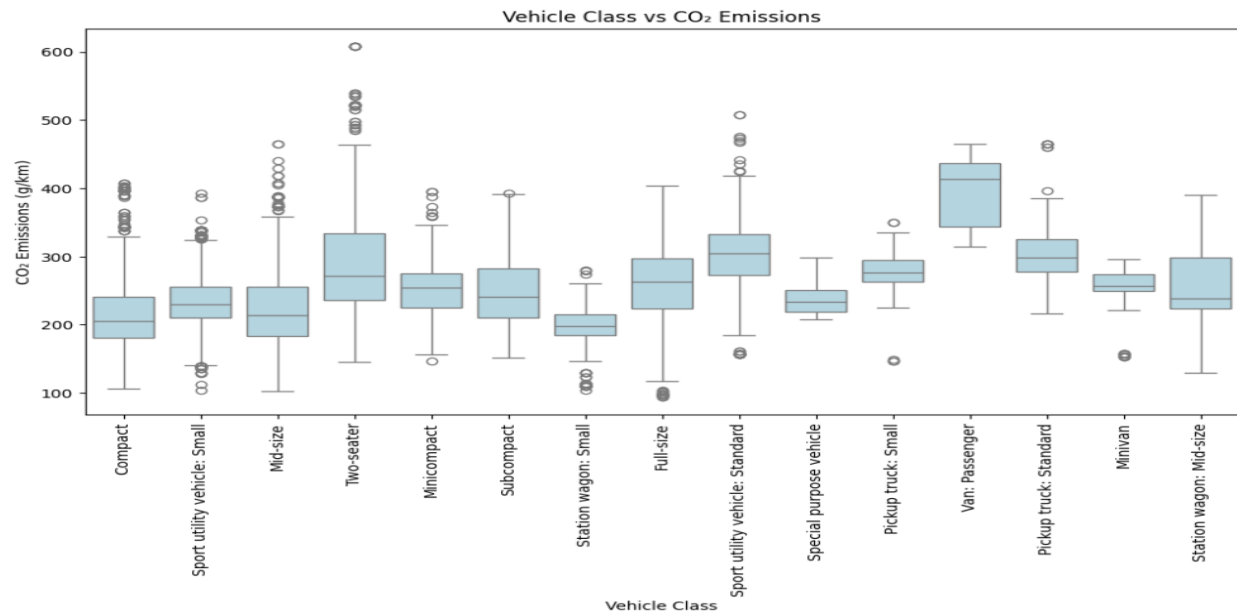


Figure 14 presents the distribution of CO₂ emissions across vehicle classes. Passenger vans have the highest median CO₂ emissions, followed by standard SUVs, standard pickup trucks, and two-seater vehicles. In contrast, compact cars and small station wagons display substantially lower median emission values. The wider spread observed for larger vehicle classes also indicates greater variation in emissions within these categories.

Stakeholder Impact

These findings can help transportation planners identify vehicle categories that contribute most to greenhouse gas emissions and prioritise strategies that encourage the adoption of lower-emission vehicle classes. The results can also support evidence-based policy development, infrastructure planning, and sustainability initiatives aimed at reducing transportation-related emissions. Consumers can use this information to compare vehicle categories when making environmentally responsible purchasing decisions and to select vehicles that better align with fuel efficiency and environmental sustainability goals.

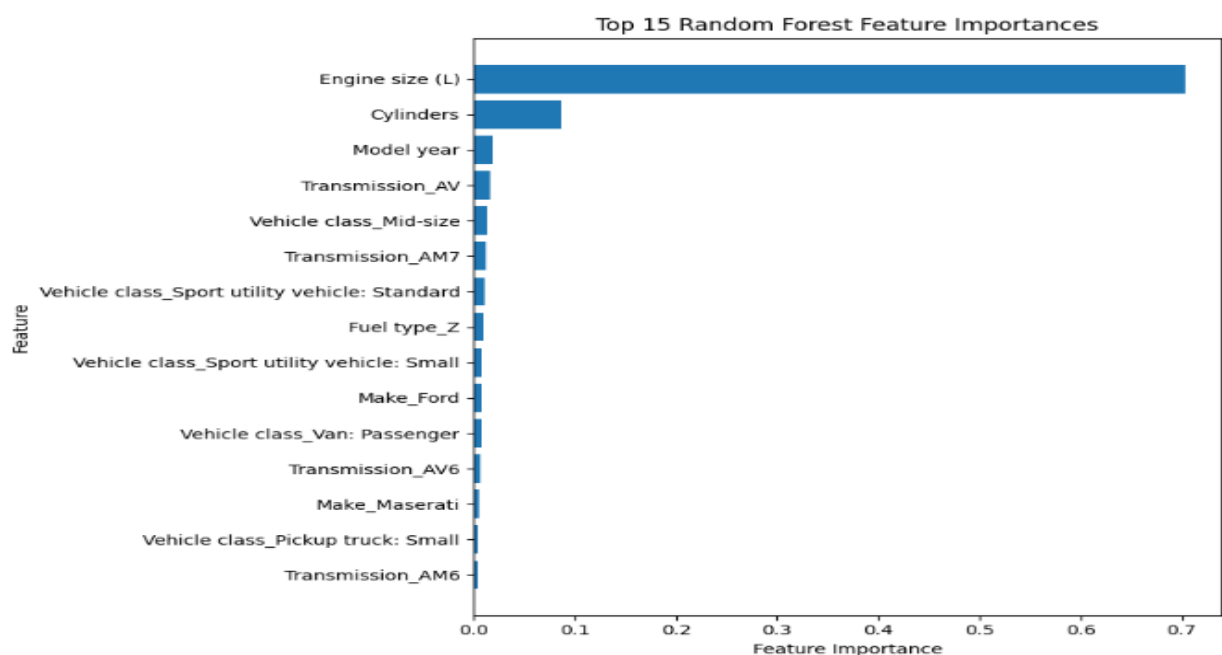
7.3 Research Question 2

What vehicle characteristics contribute the most to higher emissions in Canadian light-duty vehicles, and how can these insights encourage lower-emission vehicle choices?

The Random Forest model identified engine size as the most influential predictor of CO₂ emissions, followed by the number of cylinders. Other variables, including model year, vehicle class, transmission type, and fuel type, also contributed to the prediction, although their influence was considerably smaller as can see in figure 15. These findings indicate that vehicle design characteristics, particularly engine size and engine configuration, play the greatest role in determining CO₂ emissions.

The dominance of engine size and cylinders indicates that larger and more powerful engines are generally associated with higher CO₂ emissions. Although factors such as vehicle class and fuel type also affect emissions, their influence is secondary once engine characteristics are considered. This suggests that reducing engine displacement or improving engine efficiency could have a greater impact on lowering vehicle emissions than changes to other vehicle attributes alone.

Figure 15. Random Forest Feature Importance



Stakeholder Impact

These findings provide valuable insights for multiple stakeholders. Consumers can prioritise vehicles with smaller engines and fewer cylinders when comparing environmentally responsible options. Vehicle manufacturers can use the results to focus research and development on improving engine efficiency and reducing emissions. Policymakers and transportation

sustainability analysts can also use these findings to design regulations and incentive programs that encourage the adoption of lower-emission vehicle technologies.

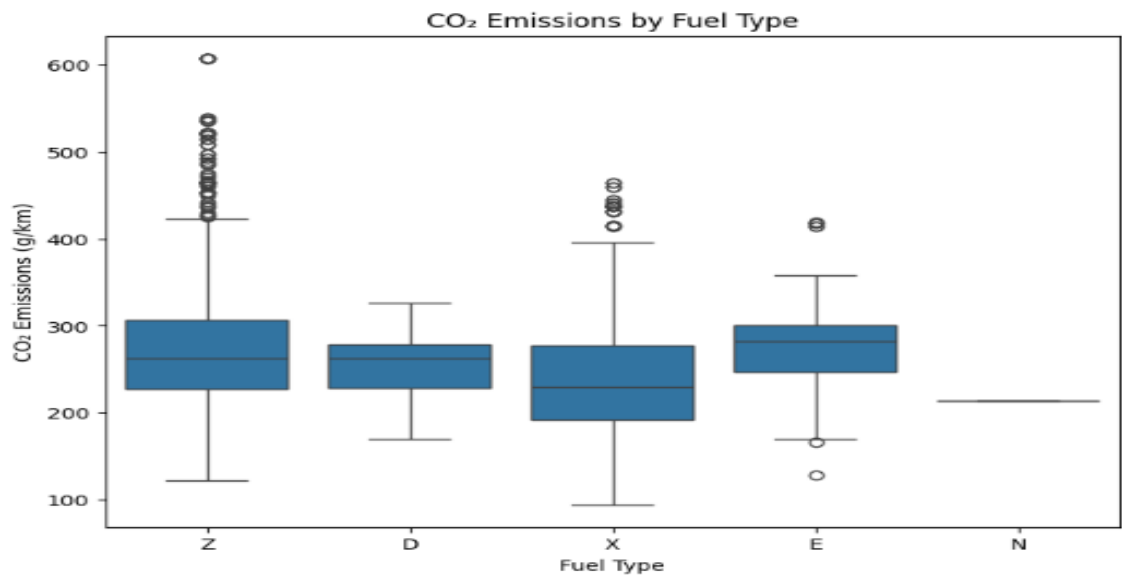
7.4 Research Question 3

How can vehicle specifications such as engine size, cylinders, and fuel type help consumers compare more environmentally responsible vehicle options?

The results indicate that vehicle specifications can provide practical guidance for comparing the environmental performance of different vehicles before purchase. In particular, fuel type influences CO₂ emissions, while the findings from the Random Forest model showed that engine size and the number of cylinders is the strongest predictors of emission levels. Together, these characteristics allow consumers to compare vehicles using readily available specifications instead of relying solely on official emission ratings.

Figure 16 below compares the distribution of CO₂ emissions across different fuel types. Vehicles using E85 ethanol (E) exhibit the highest median CO₂ emissions, while regular gasoline (X) generally has lower median emissions among the major fuel types in the dataset. Diesel (D) vehicles show relatively consistent emission levels with less variability, whereas premium gasoline (Z) displays the widest range of emissions, indicating substantial variation across vehicle models. The limited number of natural gas (N) vehicles means that observations for this fuel type should be interpreted cautiously.

Figure 16. CO₂ Emission by Fuel Type



Stakeholder Impact

These findings help consumers make more informed purchasing decisions by showing that vehicle specifications available before purchase can be used to estimate environmental impact. Manufacturers can use these insights to improve engine efficiency and develop cleaner vehicle technologies, while policymakers and transportation planners can support lower-emission vehicle adoption through targeted incentives and public awareness initiatives.

7.5 Discussion of Findings

The findings of this study are generally consistent with the literature reviewed in Section 2. Similar to previous studies, the Random Forest model achieved better predictive performance than Multiple Linear Regression, indicating that non-linear machine learning techniques are more effective for modelling vehicle CO₂ emissions. The results also confirmed that engine size and the number of cylinders are the strongest contributors to vehicle emissions, supporting earlier research on vehicle emission prediction. Unlike some previous studies, this research excluded fuel consumption and environmental rating variables from the final reported models to reduce target leakage. This provides a more realistic assessment of model performance while demonstrating that vehicle specifications alone can effectively predict CO₂ emissions.

8. Limitations and Ethics

This study has several limitations. First, the models were developed using manufacturer-reported vehicle specifications collected under standardised testing conditions, which may not fully represent real-world driving performance. Second, variables such as driving behaviour, traffic conditions, road characteristics, weather, vehicle maintenance, and vehicle age were not included because they were unavailable in the dataset. Third, the analysis was limited to Canadian light-duty vehicles from the 2015–2024 model years, so the findings may not generalise to vehicles from other countries or different time periods. Finally, although the Random Forest model achieved strong predictive performance, its predictions depend on the quality and scope of the available data.

Although the dataset is publicly available and contains no personal information, responsible use of predictive models remains important because model predictions may influence transportation planning and vehicle purchasing decisions.

9. Future Work

Future research can improve this study by incorporating additional variables that better reflect real-world driving conditions, such as traffic patterns, weather, driving behaviour, vehicle age, and maintenance history. Evaluating other machine learning algorithms, including Gradient Boosting and XGBoost, may further improve prediction accuracy. The proposed framework could also be extended to newer vehicle models, electric and hybrid vehicles, or datasets from other countries to assess its generalisability. These improvements would support the development of more robust and practical models for predicting vehicle CO₂ emissions.

10. Conclusion

This study developed and evaluated machine learning models to predict CO₂ emissions from Canadian light-duty vehicles using publicly available vehicle specifications. Multiple Linear Regression was used as the baseline model and compared with a Random Forest model. The evaluation demonstrated that the Random Forest model achieved superior predictive performance while maintaining acceptable computational efficiency and stable results across the validation procedure. The findings also identified engine size and the number of cylinders as the most influential predictors of CO₂ emissions.

The research questions proposed in Milestone 1 were answered using quantitative evidence from the developed models, feature importance analysis, and visualisations. The results provide useful insights into the factors associated with vehicle emissions and demonstrate how machine learning can support transportation sustainability through evidence-based decision making. Overall, this study achieved its objective of developing a reproducible framework for predicting vehicle CO₂ emissions while providing meaningful technical and stakeholder-focused insights.

11. GenAI Declaration

Generative Artificial Intelligence (GenAI) was used only to improve my understanding of Random Forest hyperparameter tuning during Milestone 4. The AI-assisted explanations were used for learning purposes only and to better understand the role of different hyperparameters in model development. All model implementation, parameter selection, code execution, analysis, interpretation of results, and conclusions were completed, verified, and justified by the author. The Python code was executed and validated in Google Colab to ensure that the reported results were accurate.

12. GitHub Repository and Reproducibility

The complete source code, Jupyter notebooks, project report, dataset, and supporting files used in this project are available in the GitHub repository below. The repository also includes a README.md file containing setup instructions, execution steps, project structure, and a requirements.txt file to support reproducibility. The GitHub repository includes a requirements.txt file listing the Python libraries required to reproduce the analysis.

Repository Link: <https://github.com/IndrajeetSingh121/CIND820-CO2-Emission-Prediction>

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Appendix A. Features Used in the Final Models

Component	Variables
Target Variable	CO ₂ Emissions (g/km)
Multiple Linear Regression Predictors	Model Year, Make, Vehicle Class, Engine Size (L), Cylinders, Transmission, Fuel Type
Tuned Random Forest Predictors	Model Year, Make, Vehicle Class, Engine Size (L), Cylinders, Transmission, Fuel Type
Variables Excluded	Vehicle Model, City Fuel Consumption (L/100 km), Highway Fuel Consumption (L/100 km), Combined Fuel Consumption (L/100 km), Combined Fuel Consumption (mpg), CO ₂ Rating, Smog Rating

Note: Fuel consumption variables, CO₂ Rating, and Smog Rating were excluded from the final models to reduce target leakage and provide a more realistic evaluation of model performance.